

Bayesian Estimation

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Bayesian Estimation

- Problem Definition

- Given a dataset $\mathcal{D} = \{X_1, X_2, \dots, X_n\}$ drawn from a distribution $P(X|\theta)$, i.e., $X_i \sim P, \forall i$
- Assumptions:
 - The samples are independent and identically distributed (i.i.d.)
 - P is known but not θ
- Aim: Estimate θ
- The parameters θ are modeled as random variables $\rightarrow$ incorporating all the prior knowledge we have about $\theta \rightarrow$ called the prior $p(\theta)$
- Using Bayes' Rule, the posterior is

$$p(\theta|\mathcal{D}) = \frac{p(\theta) p(\mathcal{D}|\theta)}{p(\mathcal{D})}$$

- Estimate θ based on the posterior
- We can use Maximum a Posterior (MAP) estimate as

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} p(\theta|\mathcal{D}) = \underset{\theta}{\operatorname{argmax}} p(\theta) p(\mathcal{D}|\theta)$$

Bayesian Estimation

- The estimation is simpler if we assume the likelihood, $p(\mathcal{D}|\theta)$ and the prior $p(\theta)$ are from the same family of distributions
- Then the prior is called the conjugate prior

Bayesian Estimation-Gaussian

- Problem

- Given a dataset $\mathcal{D} = \{X_1, X_2, \dots, X_n\}$ where each $X_i \sim \mathcal{N}(\mu, \sigma^2)$
- Assume that σ^2 is known and μ is unknown
- Aim is to estimate μ

- Solution

- Assume conjugate prior: $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$
- Now the posterior is

$$\begin{aligned} p(\mu|\mathcal{D}) &\propto \exp \left[-\frac{1}{2} \left(\frac{\sum_{i=1}^n X_i^2}{\sigma^2} + \frac{n\mu^2}{\sigma^2} - 2\mu \frac{\sum_{i=1}^n X_i}{\sigma^2} + \frac{\mu^2}{\sigma_0^2} + \frac{\mu_0^2}{\sigma_0^2} - 2\mu \frac{\mu_0}{\sigma_0^2} \right) \right] \\ &\propto \exp \left[-\frac{1}{2} \left(\mu^2 \left(\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right) - 2\mu \left(\frac{\sum_{i=1}^n X_i}{\sigma^2} + \frac{\mu_0}{\sigma_0^2} \right) \right) \right] \end{aligned} \quad (1)$$

Bayesian Estimation-Gaussian

- We observe that posterior $p(\mu|\mathcal{D})$ is also Gaussian
- We need to write it in the standard form to find its mean and variance parameter
- Suppose $\mu|\mathcal{D} \sim \mathcal{N}(\mu_n, \sigma_n^2)$, then

$$p(\mu|\mathcal{D}) \propto \exp \left[-\frac{1}{2} \left(\frac{\mu^2}{\sigma_n^2} - 2\mu \frac{\mu_n}{\sigma_n^2} \right) \right]$$

- Comparing (1) and the above equation,

$$\begin{aligned} \frac{1}{\sigma_n^2} &= \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \\ \frac{\mu_n}{\sigma_n^2} &= \frac{\sum_{i=1}^n X_i}{\sigma^2} + \frac{\mu_0}{\sigma_0^2} \end{aligned}$$

Bayesian Estimation-Gaussian

- By solving the above two equation we get

$$\mu_n = \frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2} \left(\frac{\sum_{i=1}^n X_i}{\sigma^2} + \frac{\mu_0}{\sigma_0^2} \right)$$

- Denoting $\bar{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$ and simplifying the above equation yields,

$$\mu_n = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \bar{\mu} + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \mu_0 \quad (2)$$

- Convex combination of prior knowledge and the information from observations
- The MAP estimate is μ_n since Gaussian is maximum at its mean
- As $n \rightarrow \infty$, $\mu_n \rightarrow \bar{\mu}$ and $n \rightarrow 0$, $\mu_n \rightarrow \mu_0$